

Cryptography

Unit- (2.5) Public Key Cryptosystem

Drawbacks of Symmetric Key System

1. Key Distribution :

One of the major challenge in symmetric key cryptography is securely distributing the shared secret key among all parties who need to communicate. If a large number of users or devices need to communicate securely, securely distributing and managing the keys becomes complex and can be prone to security risk.

2. Lack of Scalability :

As the number the number of communication parties increases, the number of key required for secure communication increases exponentially with symmetric key cryptography. For example, in a network with N- participants, $N(N-1)/2$ keys would be needed. This can quickly become unwieldy and difficult to manage in large scale systems.

Drawbacks of Symmetric Key System

3. Lack of Forward Secrecy :

In symmetric key system, if the shared secret key is compromised or leaked, all past and future communications encrypted with that key are also compromised.

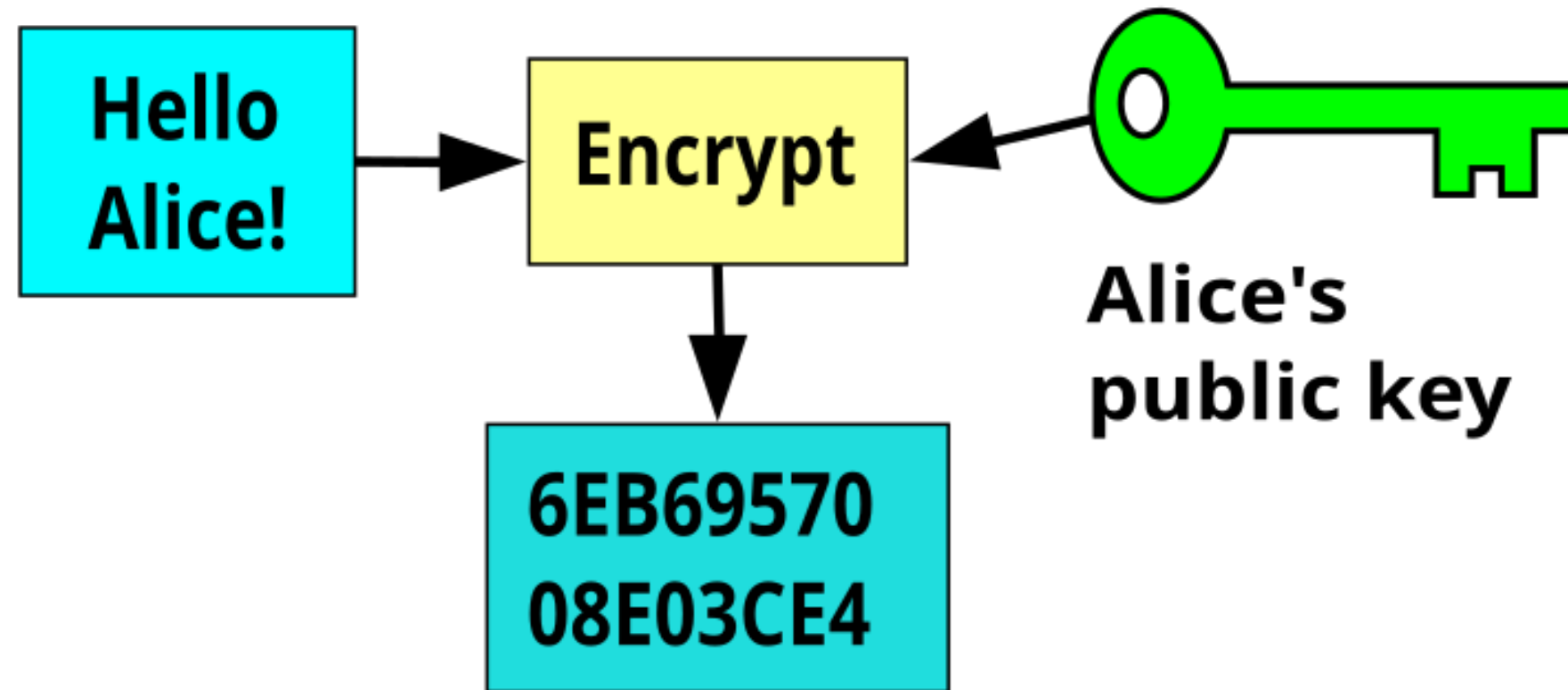
4. Limited Authentication :

Symmetric key systems do not provide inherent authentication of the communicating parties.

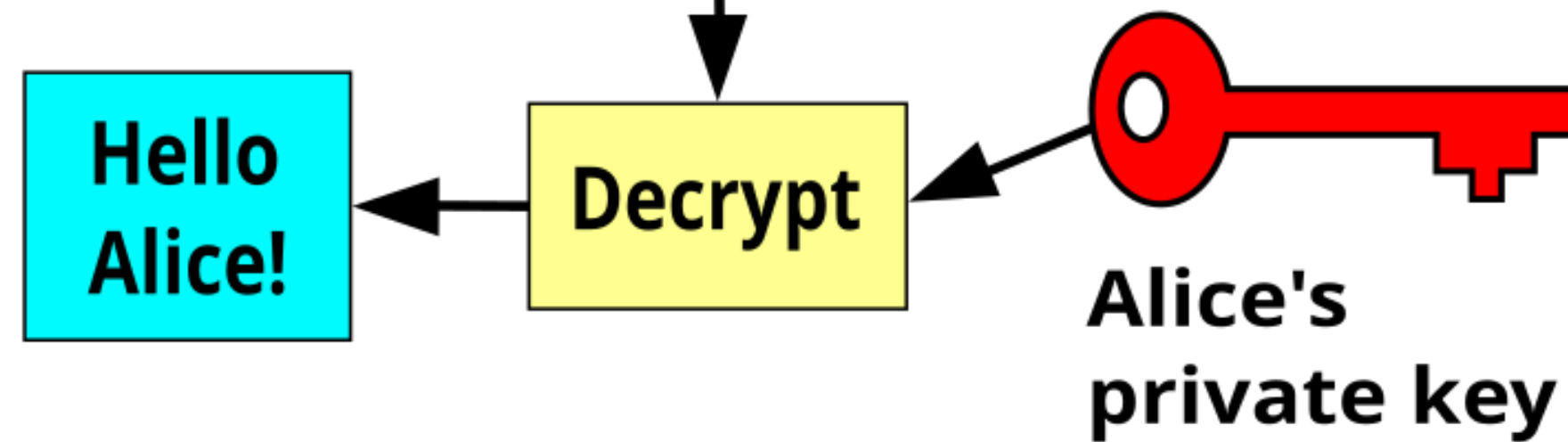
Public Key Cryptography

- Public key cryptography, also known as asymmetric cryptography, is a cryptographic scheme that uses a pair of keys – a public key and a private key for secure communication and information exchange.
- The basic idea behind public key cryptography is that the public key can be openly shared, while the private key must be kept secret.
- The public key is used to encrypt data or verify data signatures, while the private key is used to decrypt the encrypted data or create a digital signatures. This enable secure communication and authentication without the need for shared – secret key.

Bob



Alice



Application of Public Key System

- Public-key systems are characterized by the use of a cryptographic algorithm with two keys, one held private and one available publicly. Depending on the application, the sender uses either the sender's private key or the receiver's public key, or both, to perform some type of cryptographic function. In broad terms, we can classify the use of public-key cryptosystems into three categories:

1. Encryption/Decryption

- The sender encrypts a message with the recipient's public key.

2. Digital Signature

- The sender “sign” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.

3. Key Exchange

- Two sides cooperate to exchange secret key, session key.
- Several different approaches are possible, involving the private key(s) of one or both parties.

Distribution of Public Keys

- The public key can be distributed in four different ways-

1. Public Announcement

- Here the public key is broadcasted to every one.
- The major weakness of this method is a forgery. Any one can create a key claiming to be some one else and broadcast it.

2. Public Available Directory

- Here the public key is stored in a public directory. Public directories are trusted here, with properties like participant Registration, access and allow to modify values at any time, contains entries like Name, public-key. It is still vulnerable to forgery or tempering.

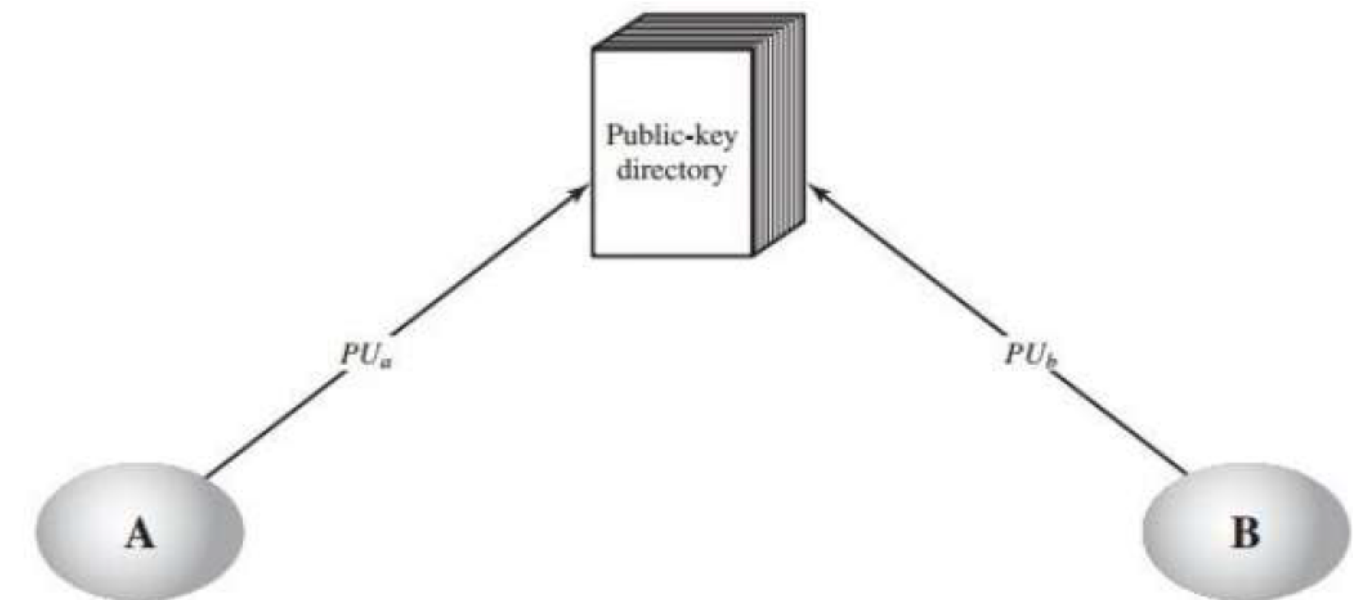
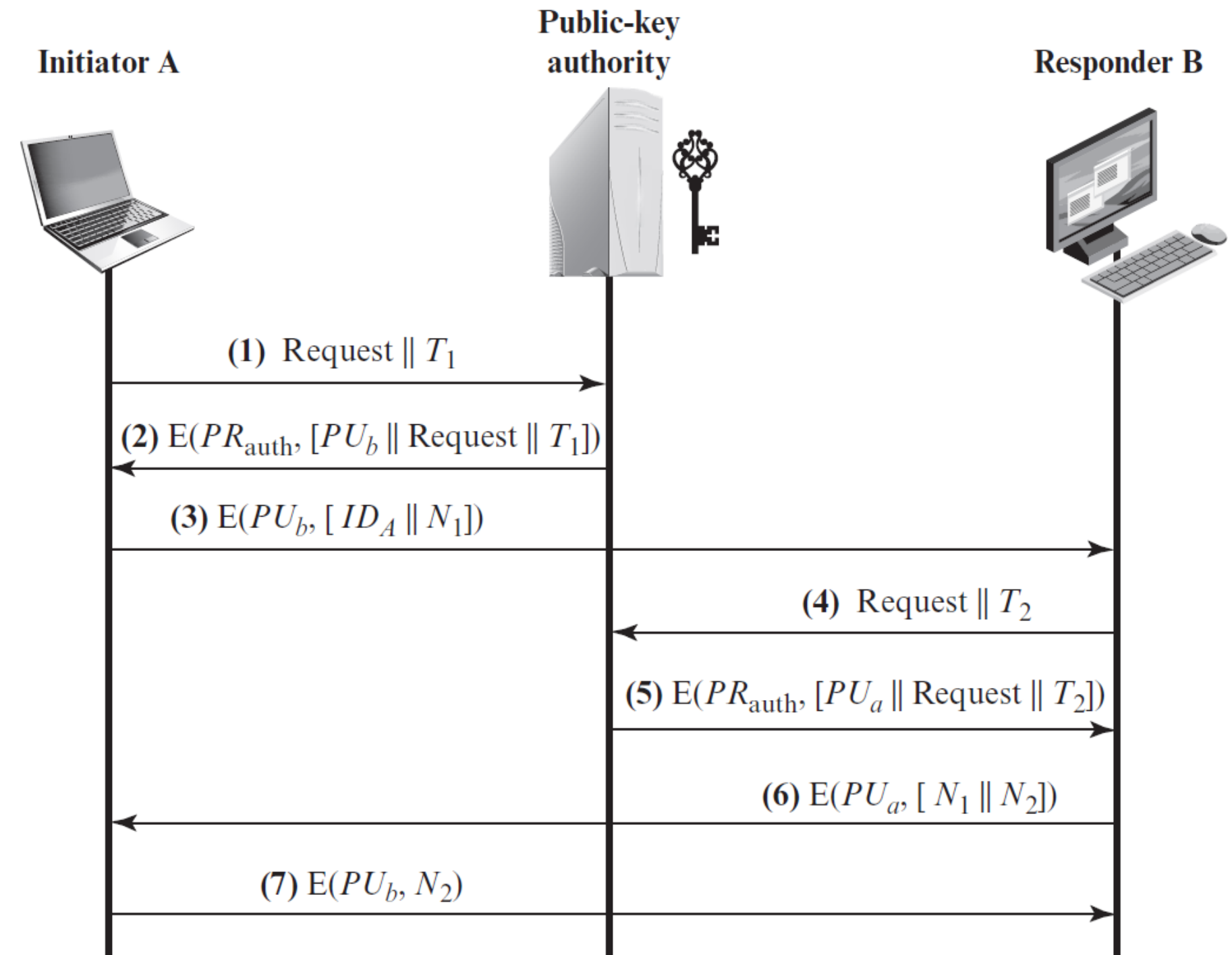


Fig: Public-key publication

Distribution of Public Keys

3. Public Key Authority

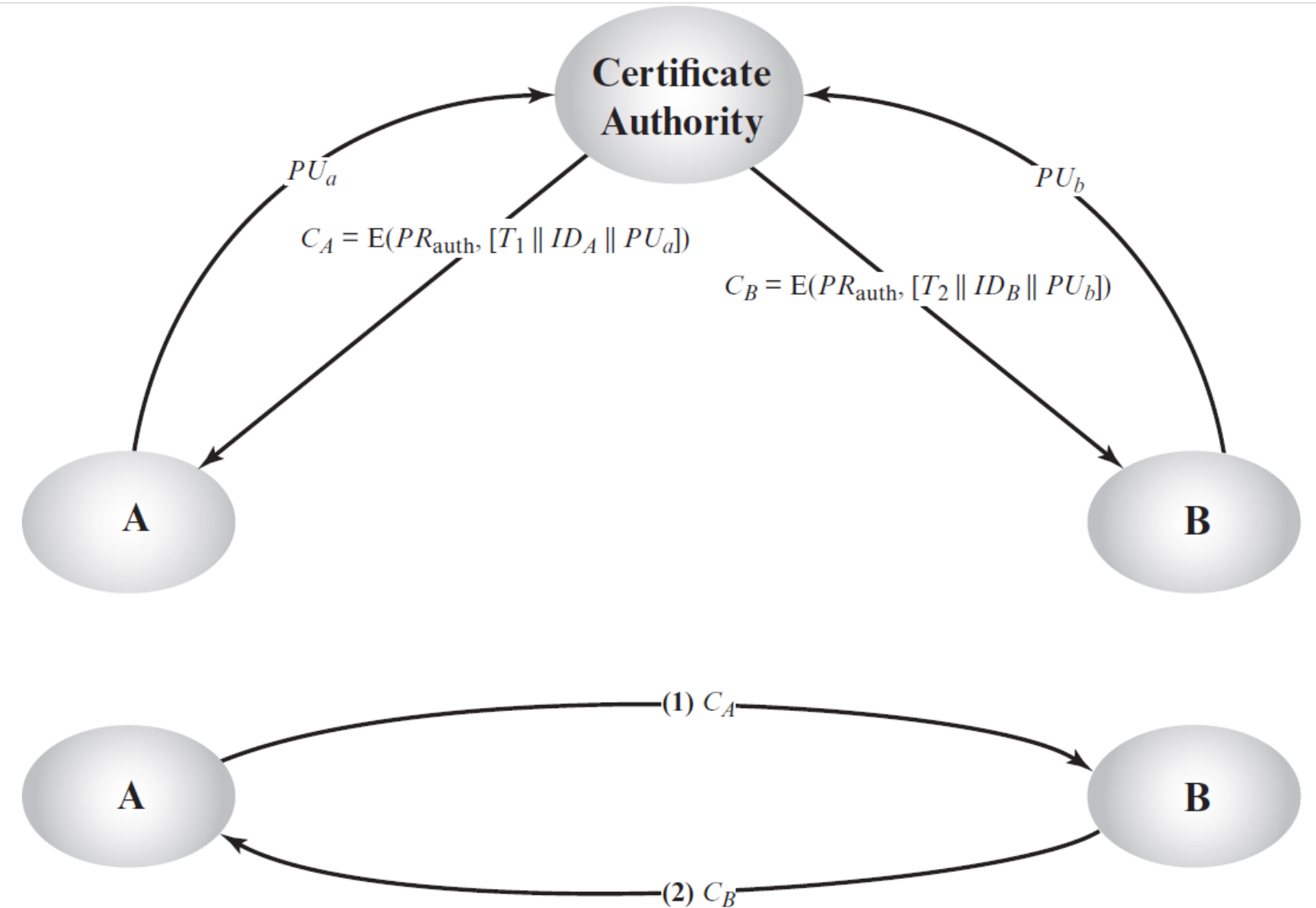
- It is similar to directory but, improves security by tightening control over the distribution of keys from the directory. It requires user to know the public key for the directory. Whenever the keys are needed, real-time access to directory is made by the user to obtain any desired public key securely.



Distribution of Public Keys

4. Public Certification :

- This time authority provides a certificate which binds identity to the public key to allow key exchange without real time access to the public authority each time. The certificate is accompanied by some other information such as period of validity, right of use etc. All of this content is signed by the trusted public-key or certificate Authority (CA) and it can be verified by anyone possessing the authority's public-key.



Diffie-Hellman Key Exchange

- The Diffie-Hellman key exchange is a method for securely exchanging cryptographic keys over an insecure channel. It is a fundamental building block for many secure communication protocols.

1. Global public element :

- $q \rightarrow$ prime number
- $\alpha \rightarrow \alpha < q$ & α is primitive root of q

2. User A key generation :

- Select private X_A s.t $X_A < q$
- Calculate public Y_A s.t $Y_A = \alpha^{X_A} \bmod q$

3. User B key generation :

- Select private X_B s.t $X_B < q$
- Calculate public Y_B s.t $Y_B = \alpha^{X_B} \bmod q$

Diffie-Hellman Key Exchange

4. Calculation of secret key by User A :

$$K = (Y_B)^{X_A} \bmod q$$

5. Calculation of secret key by User B :

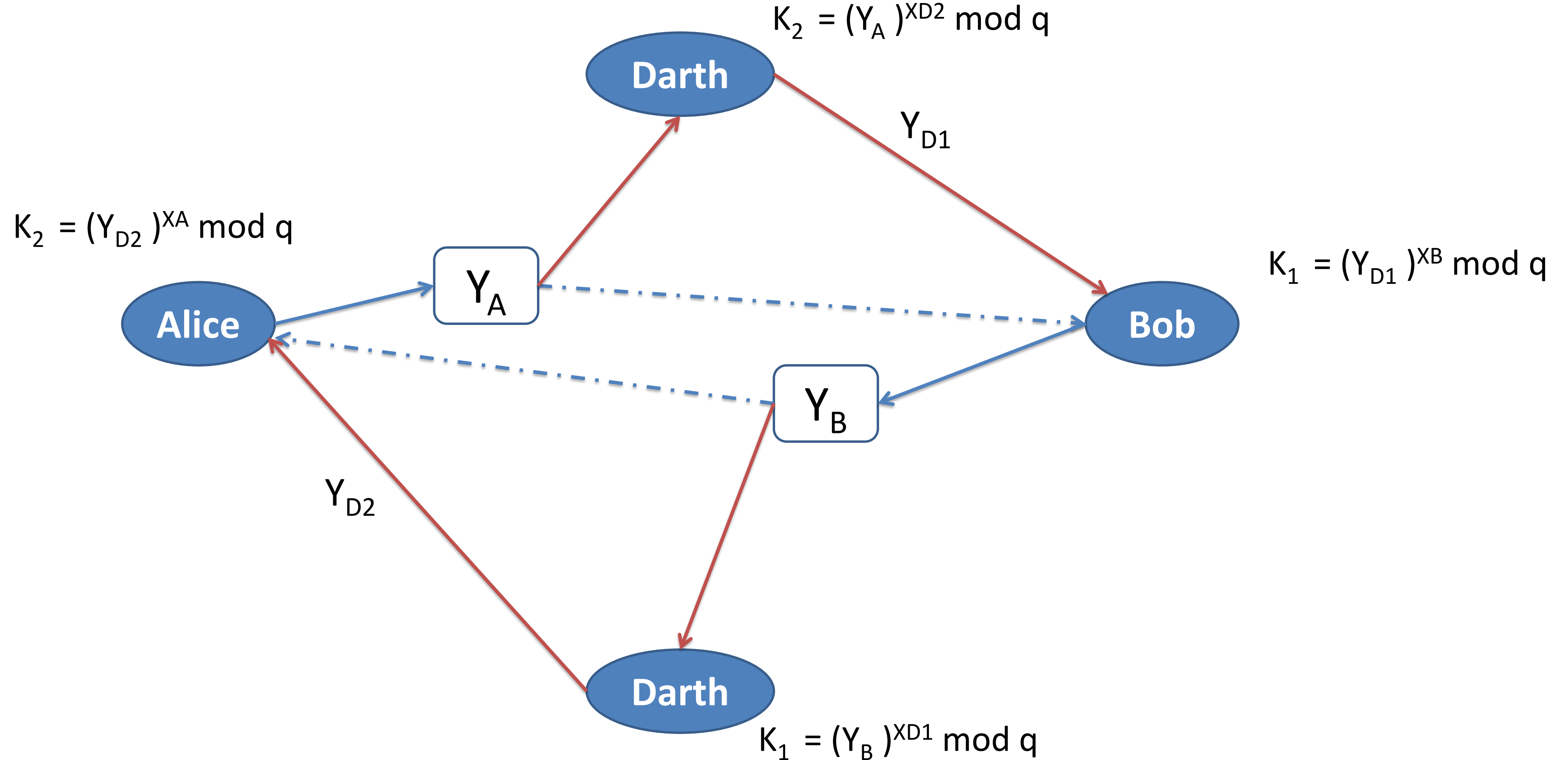
$$K = (Y_A)^{X_B} \bmod q$$

Example

- $q = 13$
- $\alpha = 2$
- $X_A = 3$ & $X_B = 5$

- $Y_A = \alpha^{X_A} \bmod q$
 $= 2^3 \bmod 13$
 $= 8$
- $Y_B = \alpha^{X_B} \bmod q$
 $= 2^5 \bmod 13$
 $= 6$
- 'A' Compute key $K = (Y_B)^{X_A} \bmod q = 6^3 \bmod 13 = 8$
- 'B' Compute key $K = (Y_A)^{X_B} \bmod q = 8^5 \bmod 13 = 8$

Man-in-the-Middle Attack (D-H)



Man-in-the-Middle Attack (D-H)

- Darth prepare for the attack by generating two random private keys X_{D1} & X_{D2} and then computing the corresponding public keys Y_{D1} & Y_{D2} .
- Alice transmit Y_A to bob. Darth intercept Y_A and transmit Y_{D1} to bob. Darth also calculate $K_2 = (Y_A)^{X_{D2}} \bmod q$.
- Bob receive Y_{D1} and calculate $K_1 = (Y_{D1})^{X_B} \bmod q$.
- Bob transmit Y_B to Alice. Darth intercept Y_B and transmit Y_{D2} to Alice. Darth also calculate $K_1 = (Y_B)^{X_{D1}} \bmod q$.
- Alice receive Y_{D2} and calculates $K_2 = (Y_{D2})^{X_A} \bmod q$.
- At this point Bob and Alice think that they share secret key but instead Bob and Darth share secret key K_1 & Alice and Darth share secret key K_2 .

RSA Algorithm

- RSA algorithm is a widely used public key cryptography algorithm that was invented by Ron Rivest, Adi Shamir, and Lenoard Adleman in 1978 AD.
- It is named after their initials (Rivest, Shamir, Adleman).
- The RSA algorithm is widely used for Secure communication, Digital Signature, & Encryption of data.
- The working of RSA algorithm is as follows :

RSA Algorithm

1. Key Generation :

- Select p, q where, p & q both prime & $p \neq q$
- Calculate $n = p * q$
- Calculate $\phi(n) = (p-1) * (q-1)$
- Select integer e s.t. $\gcd(\phi(n), e) = 1 ; 1 < e < \phi(n)$
- Calculate d i.e $d = e^{-1} \pmod{\phi(n)}$
- Public key $PU = \{ e, n \}$
- Private key $PR = \{ d, n \}$

2. Encryption :

- Plaintext $m < n$
- Ciphertext $C = m^e \pmod n$

3. Decryption :

- Ciphertext C
- Plaintext $M = C^d \pmod n$

RSA Algorithm

Example

$$p = 5 \text{ \& } q = 19$$

$$n = pq = 5 * 19 = 95$$

$$\phi(n) = (5 - 1)(19 - 1) = 4 * 18 = 72$$

Choose e , such that $1 < e < 72$ and co-prime to 72.

$$\therefore e = 5$$

Calculate d by using;

$$ed \equiv 1(\text{mod } \phi(n))$$

$$5 * d \equiv 1(\text{mod } 72)$$

$$5 * 29 \equiv 1(\text{mod } 72)$$

$$\therefore d = 29$$

So, the public key is $\{e, n\} = \{5, 95\}$ and the private key is $\{d, n\} = \{29, 95\}$.

Consider $m = 19$

Encryption:

$$\begin{aligned} c &= m^e \text{ mod } n \\ &= 19^5 \text{ mod } 95 \\ &= 19 \end{aligned}$$

Decryption:

$$\begin{aligned} m &= c^d \text{ mod } n \\ &= 19^{29} \text{ mod } 95 \\ &= 19 \end{aligned}$$

!!!Thank you!!!